

UE22CS320B – Capstone Project Phase – II - Weekly Status Report

Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	06/01/2025	Week Ending Date	12/01/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Extensive Literature Survey	06/01/2025	
2	Design analysis started for implementation of the Project	08/01/2025	
In Progress			
#	Tasks	Date	Remarks
1	A draft of the architecture diagram was being made.	10/01/2025	
2	Looking for potential tools to help us implement the required functionalities.	10/01/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Finalize the architecture diagram	12/01/2025	
2	Tools to be used for implementing the project	12/01/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	13/01/2025	Week Ending Date	19/01/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Draft Architecture Diagram	13/01/2025	
2	Basic Tool research for AVIOS	16/01/2025	
In Progress			
#	Tasks	Date	Remarks
1	The Final Architecture Diagram was decided.	18/01/2025	
2	Selected tools like scikit-learn, Perf, psutil, Tensor and others	18/01/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Testing the tools	19/01/2025	
2	Identification of key modules required for implementation	19/01/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	20/01/2025	Week Ending Date	26/01/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	The Architecture Diagram	20/01/2025	
2	Identification of Potential Tools.	23/01/2025	
In Progress			
#	Tasks	Date	Remarks
1	Identified system requirements like CPU, memory, and other	25/01/2025	
2	Scheduling task to be made dynamic using ai predictions	25/01/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Identification of key modules required for implementation	26/01/2025	
2	Dependencies and Assumptions	26/01/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	27/01/2025	Week Ending Date	02/02/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Conducted background study on Linux CFS and identified gaps	27/01/2025	
2	Identified data flow and kernel interaction	29/01/2025	
In Progress			
#	Tasks	Date	Remarks
1	Key modules which are required for smooth implementation	30/01/2025	
2	Dependencies and Assumptions	30/01/2025	
3	Choose Hybrid Monolithic and Modular Design Approach for implementation	30/01/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Find ML models which [it to our use cases	02/02/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	03/02/2025	Week Ending Date	09/02/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Key modules which are required for smooth implementation	03/02/2025	
2	Design Approaches and Principles	06/02/2025	
In Progress			
#	Tasks	Date	Remarks
1	SVM, GAN, Random Forest, Xgboost, MLP, Random Forest	08/02/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Research for more similar Datasets for the ML Models	09/02/2025	
2	Train the models on the datasets	09/02/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	10/02/2025	Week Ending Date	16/02/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	ML models which will be used in the projects for analysis.	10/02/2025	
2	Tools used in reporting	13/02/2025	
In Progress			
#	Tasks	Date	Remarks
1	Finding relevant datasets for each of the different models	14/02/2025	
2	Deciding which models can be used for which analysis	14/02/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Extract Data from Ubuntu Systems from College Labs	16/02/2025	
2	Cleaning up data for validation	16/02/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	17/02/2025	Week Ending Date	23/02/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Searching Datasets for ML Models	15/02/2025	
In Progress			
#	Tasks	Date	Remarks
1	Continuing the data extraction from the Ubuntu Systems	17/02/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Pre-processing the Data for giving them as input to ML Models	24/02/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	24/02/2025	Week Ending Date	02/03/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Data Extraction using psutil tools	23/02/2025	
2	Identified the importance of specific attribute	23/02/2025	
In Progress			
#	Tasks	Date	Remarks
1	Pre-process the data which was extracted from the mobile.	25/02/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Further Pre-processing of the Extracted Data	03/03/2025	

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Batch No	144		
Week Starting Date	03/03/2025	Week Ending Date	09/03/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Pre-process the data which was extracted from the Ubuntu Lab System	02/03/2025	
In Progress			
#	Tasks	Date	Remarks
1	Generating a static Dataset using GAN model from the pre processed clean dataset.	03/03/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Start training the ML models using this dataset	10/03/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	10/03/2025	Week Ending Date	16/03/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Trained various ML models for classifying the task	07/03/2025	
2	Achieved training the models with average accuracy among all models with 90% after cross validating.	09/03/2025	
In Progress			
#	Tasks	Date	Remarks
1	Improving the accuracy of the models already trained	10/03/2025	
2	Finding other methods to train the model	10/03/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Research and Training the Model	17/03/2025	
2	Improving the accuracy and input features of the model	19/03/2025	

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Batch No	144		
Week Starting Date	17/03/2025	Week Ending Date	23/03/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Understood different other models that can be used for this project.	16/03/2025	
2	Improved the accuracy by 2.632%	16/03/2025	
In Progress			
#	Tasks	Date	Remarks
1	Trying to improve accuracy	17/03/2025	
2	Finding more models and how to integrate into real time testing	17/03/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Improve the input and output features and methods of the model	24/03/2025	
2	Improve the Accuracy and precision of the model	24/03/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	24/03/2025	Week Ending Date	30/03/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Prepare Models	23/03/2025	
2	Train the models with the pre found datasets.	23/03/2025	
In Progress			
#	Tasks	Date	Remarks
1	Improve the Accuracy and Features	24/03/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Improve the Accuracy of already trained models	31/03/2025	

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Project Title	AVIOS – Adaptive Vigilant Intelligent Operating System		
Batch No	144		
Week Starting Date	31/03/2025	Week Ending Date	06/04/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331

Activities Completed

#	Tasks	Date	Remarks
1	Improve the Accuracy of the Models. Such as random forest, Xgboost, svm	30/03/2025	
2	Improved model accuracy to 93.67%	30/03/2025	

In Progress

#	Tasks	Date	Remarks
1	Better method of encoding and handling missing value efficiently	31/03/2025	

Plan for Next Week

#	Tasks	Date	Remarks
1	Better ways of handling missing values	07/04/2025	

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Batch No	144		
Week Starting Date	07/04/2025	Week Ending Date	13/04/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	The missing values are handled more efficient manner	06/04/2025	
2	Encoding has been achieved	06/04/2025	
In Progress			
#	Tasks	Date	Remarks
1	Prepare for performing feature analysis through feature engineering	07/04/2025	
2	Enhance the data visualization of the feature for better understanding	07/04/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Complete the training of the first part of project	14/04/2025	

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Batch No	144		
Week Starting	14/04/2025	Week Ending Date	20/04/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Comparing the feature/task type activities of the system	13/04/2025	
2	Identifying the redundancy of the task type through heat map	13/04/2025	
In Progress			
#	Tasks	Date	Remarks
1	Collected the results of the task type activities from the visualization	14/04/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Wrapping up with the model training part	21/04/2025	

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Batch No	144		
Week Starting Date	21/04/2025	Week Ending Date	27/04/2025
Name	Kurumeti Prajval Manoj Kumar KS Dhanushree K Nagaratna	SRN	PES2UG22CS274 PES2UG22CS302 PES2UG22CS176 PES2UG22CS331
Activities Completed			
#	Tasks	Date	Remarks
1	Finished with training of model that classify the task type and achieved a accuracy of 95.67% after cross validation	20/04/2025	
In Progress			
#	Tasks	Date	Remarks
1	Documenting the Results achieved.	21/04/2025	
Plan for Next Week			
#	Tasks	Date	Remarks
1	Start implementing with next part of the project predicting the accurate time slice	28/04/2025	

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